A comment on global bird abundance estimations by Callaghan et al. 2021

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The publication of Callaghan et al. (2021, Science) attempts to estimate global bird abundance by integrating expert-derived population estimates with massive citizen-science datasets. It uses a hierarchical Bayesian model to link eBird-derived relative abundance to true density for a training set of 724 species, then propagating that relationship with uncertainty across approximately 9700 species worldwide. The authors' approach to data integration, multiple imputation, and uncertainty propagation is methodologically sophisticated, and their major findings, a strong left-skew in the global species abundance distribution (SAD) and an estimated approximate 50 billion individual birds, seem quite sound. However, as with any global-scale synthesis, the analysis rests on several assumptions that warrant a second glance. Two aspects in particular motivated our reanalysis: firstly, the treatment of extremely rare species in the training data, and secondly the assumption that a fixed -1 correlation between slope and intercept in the species-specific abundance-density relationships. Both choices have meaningful implications for the stability and realism of the resulting abundance estimates.

The authors acknowledge a limitation in their study, namely that some species in the training dataset are so rarely observed that their population estimates are inherently unreliable. These species contribute disproportionately noisy information to the abundance-density model, yet their original analysis does not attempt to mitigate this problem. Given that the training model is the basis for the entire analysis, biased or unstable estimates at this early stage can propagate through the entire pipeline. Reducing the training data by removing 15 percent of the least observed species, would reduce this problem, by removing the bias produced by rarely observed species. However when looking at the results (Figure 1) it becomes clear that there is no big diversion from the original output. This indicates that the bias created by rarely observed species is neglectable and that their inclusion does not meaningfully distort the global results..

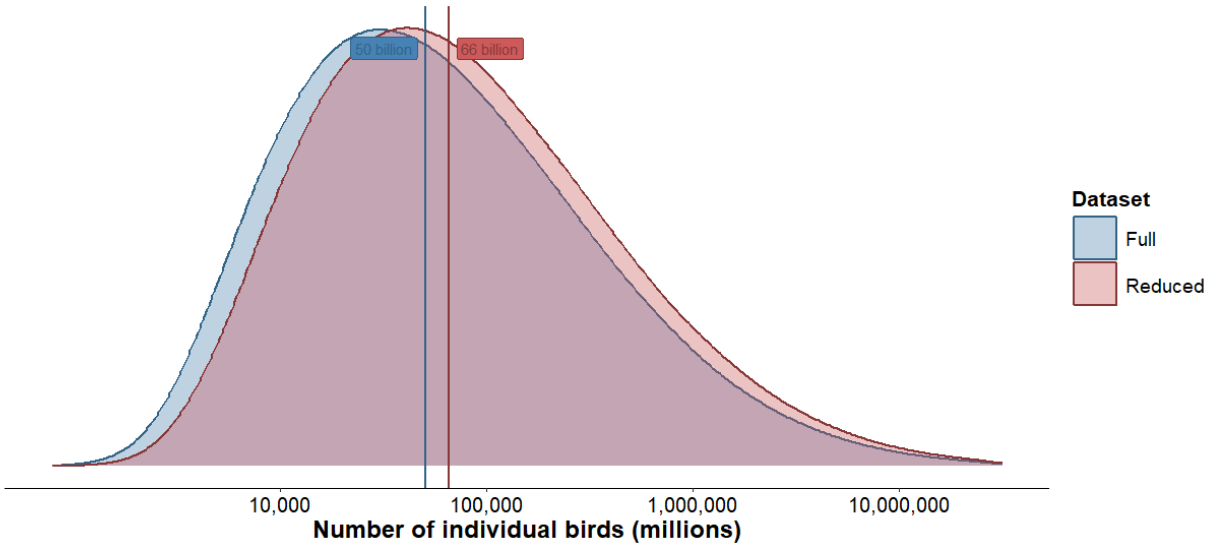

Figure 1: Comparison of abundance distributions with original dataset (full) and with reduced dataset (15% of least observed species removed).

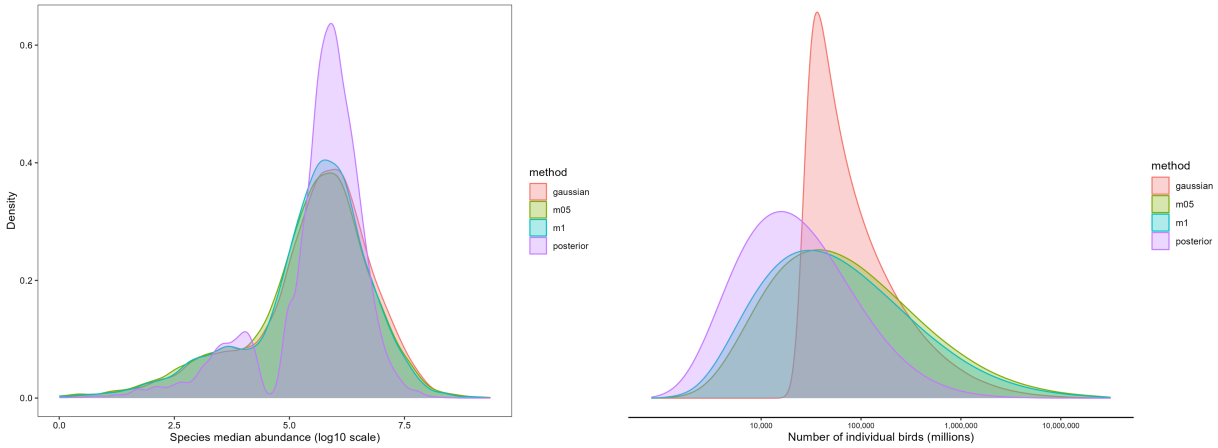

Figure 2. Species median abundance density curves along the log10 scale found on the left, and global bird abundance distributions found on the right. Both show the results for each method.

A second major assumption in Callaghan et al., was that the posterior correlation between the slope and intercept of each species' abundance-density relationship is fixed at -1. This assumption simplifies uncertainty propagation but represents the most conservative possible correlation structure. We explored three alternative approaches: a fixed –0.5 correlation, a Gaussian method that simulates slope–intercept pairs from the model's estimated variance–covariance matrix, and a posterior-derived method that uses each species' actual posterior covariance structure. At the species level, the median abundance distributions for the Gaussian, -1, and -0.5 scenarios were nearly identical, with the posterior having less variability among the species medians, given the narrower and more peaked curve. At the global scale, the gaussian method appears overconfident and strongly constrained, while the original -1 and new -0.5 method are nearly identical. The posterior-derived method, however, produced a left-shifted and narrower global abundance distribution, resulting in systematically lower total abundance estimates (Figure 2).

In summary, Callaghan et al. (2021) presents a rigorous method that advances our understanding of global bird abundance. Their integration of expert knowledge with citizen-science data is methodologically sound, and many of their main findings hold up when tested in different ways. Our reanalysis does show, however, that the way uncertainty is handled, especially the assumption about how the slope and intercept are correlated, can noticeably affect the final global abundance estimates. The influence of rarely observed species in the training data seems small, but using different, more realistic ways of modelling parameter uncertainty leads to different total abundance values. Because of this, future global abundance studies would benefit from looking more closely at how uncertainty is propagated through the analysis, since these choices can meaningfully shape the final numbers and how they are interpreted in conservation and ecological contexts.

References:

C.T. Callaghan, S. Nakagawa, & W.K. Cornwell, Global abundance estimates for 9,700 bird species. Proc. Natl. Acad. Sci. U.S.A., 118, e2023170118 (2021). https://doi.org/10.1073/pnas.2023170118.